# CHLOÉ BARJOU-DELAYRE

**Main area of research focuses on the study of cosmic expansion anisotropy with type Ia supernovae**

@ chloe.barjou-delayre@clermont.in2p3.fr  Laboratoire de Physique de Clermont Auvergne, France
https://chloebarjoudelayre.github.io/

## EDUCATION

### University

**PhD on the study of cosmic expansion anisotropy with type Ia supernovae**

October 2023-October 2026  University Clermont Auvergne, France

- Study $H_0$ anisotropy from Zwicky Transient Facility (ZTF) survey.
- Calibrate synthetic filter model of ZTF
- Analyze the systematics of the Traveling Colimated Beam Projector (TCBP)
- Member of the laboratory's seminar organizing team
- Supervised by Philippe Rosnet and co supervised by Corentin Ravoux.

**Master's Degree**

September 2021-June 2023  University Toulouse III Paul Sabatier, France

- Second year: ASEP-Astrophysics, Space Science and Planetary Science.
- First year: SUTS-Science of the Universe and Space technology.

**TESS - Toulouse graduate school of earth and space science, 30 extra ECTS in 2 years**

September 2021-June 2023  University Toulouse III Paul Sabatier, France

- A transdisciplinary complementary Master's program where a selected student follows content focused on practical experience in taking and processing data relating to Earth and Space sciences, within the wider context of social and economic implications. Courses in addition to the master's degree.

**Bachelor's Degree of physics**

September 2020-May 2021  University Toulouse III Paul Sabatier, France

**CUPGE- University Preparatory Cycle for Engineering schools**

September 2018-May 2020  University Toulouse III Paul Sabatier, France

- Two-year multidisciplinary undergraduate program to prepare for the competitive entrance exams for engineering school.

### Internships

**Are the laws of physics universal ?**

January 2023-July 2023  CAUP Portugal

- Study of the variation of the fine-structure constant. By developing an MCMC code to constrain parameters of the Davoudiasl Giardino model.
- Supervised by Dr. Carlos Martins.

**Constraining the nature of ultra-luminous X-Ray sources (ULXs)**

July 2022  IRAP Toulouse

- Using spectra we determined the luminosity and hardness ratio of ULXs in order to plot the Luminosity-Hardness diagram.
- Supervised by Dr. Olivier Godet.

## TALK, POSTER, SEMINARS, SCHOOL

**ZTF Cosmo meeting**

September 2026  Paris, France

- Talk : *Update study of anisotropic cosmic expansion with ZTF type Ia supernovae simulations*

**60th Rencontre de Moriond**

March 2026  La Tuile, Italy

- Selected poster : *Study of anisotropic cosmic expansion with ZTF type Ia supernovae simulations*

**Cosmology seminar**

Febuary 2026  Oxford, UK

- Invited talk for the cosmology seminar: *Study of cosmic expansion anisotropy with type Ia supernovae from ZTF simulations*

**Cosmology seminar**

January 2026  Birmingham, UK

- Invited talk for the cosmology seminar: *Study of cosmic expansion anisotropy with type Ia supernovae from ZTF simulations*

**Action Dark Energy (ADE)**

November 2025  Montpelier, France

- Selected talk for supernovae session : *Study of cosmic expansion anisotropy with type Ia supernovae*

**Conference Dark Side of the Universe (DSU)**

July 2025  Montreal, Canada

- Selected talk for supernovae session : *Study of cosmic expansion anisotropy with type Ia supernovae*

**ZTF calibration meeting**

March 2025  Paris, France

- Talk : *Synthetic filters calibration with Gaia spectrum.*

**ZTF Cosmology meeting**

December 2024  Barcelona, Spain

- Talk : *Study of cosmic expansion anisotropy with type Ia supernovae*

**Action Dark Energy (ADE)**

November 2024  Paris, France

- Selected talk : *Study of cosmic expansion anisotropy with type Ia supernovae*

# OUTREACH

### Public Conference

7 August 2025 — Vulcania, France

- A public conference of one hour about *'The conquest of space, the dream come true'* at the Cinemax, a room of 400 people, at the theme park Vulcania.
- https://www.vulcania.com/actualites/astronomiales/

### Host at Nuées Ardentes

14 June 2025 — Puy de Dome, France

- Animator of the physical stand at the festival Nuée Ardente. Introduction to cosmology for the general public: thanks virtual headsets that reproduce the cosmic web.
- https://lesnueesardentes.uca.fr/

### Introduction to cosmology for high school student

January 2025 — Clermont-Ferrand,France

- One hour and a half of introduction to cosmology for high school student.

### Speech-searching

January 2024 — Clermont-Ferrand,France

- Two hours of speech-searching during the international woman of science day (UNESCO).

# TELESCOPE OBSERVATION

### Traveling Collimated Beam Projector (TCBP) campaign at ZTF

May 2025 — Palomar Observatory, USA

- Two weeks of measurement of the Zwicky Transient Facility camera with the TCBP. The TCBP emit a parallel monochromatic light beam with a controllable wavelength, which allows simulating an infinite light source. That allows to measure the transmission of the optics of the telescope and the CCD quantum efficiency with and without the filters.

### Pic du Midi Observatory

July 2022 — Pyrénée, France

- 2 week of observation with the T50 for a master degree project.

### ZTF meeting

June 2024 — Stockholm, Sweden

- Talk : *Study of cosmic expansion anisotropy with type Ia supernovae and The impact of Synthetic filters on SN Ia ZTF light-curves*

### 58th Rencontre de Moriond

April 2024 — La Tuile, Italy

- Selected poster : *The impact of synthetic filters on SN Ia ZTF light-curves*

### Rodolphe Clédassou summer school

August 2023-2025 — France

- 3-year program, with two weeks of cosmology and technical lectures each year, associated to Euclid France. One talk given of 5mn.

# GRANTS

### PhD Student Research Conference

May 2026 — Clermont-Ferrand, France

- Award for Best Presentation, 200€.

### Doctoral Contract

October 2023 — Clermont-Ferrand, France

- 3-years PhD grant, $\approx 100$k€

# TEACHING (84H)

- Radioactivity lesson (2023-2025) for first year of bachelor student. Eight lessons of one hour and half for different group.
- Practical lesson of electricity (2023-2025) for first year of bachelor student. Four lessons of four hours.

# REFEREES

**Prof. Philippe Rosnet**
@ Laboratoire de Physique de Clermont d'Auvergne (LPCA)
philippe.rosnet@clermont.in2p3.fr
PhD supervisor

**Dr. Nicolas Regnault**
@ Laboratoire Physique Nucléaire et Hautes Energies (LPNHE)
nicolas.regnault@lpnhe.in2p3.fr

**Dr. Benjamin Raccine**
@ Centre de Physique des Particules de Marseille (CPPM) and Department of Astronomy at Yale University
racine@cppm.in2p3.fr